

Murphy Diggins

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EDUCATION

University of Wisconsin-Madison

Madison, WI

B.S. Biomedical Engineering, Biomaterials Track. Involvement: Phi Delta Theta (Member, Judicial Board), Biomedical Engineering Society, IFC Softball, Party of Poets, Humorology.
GPA: 3.50/4.00.

Dec. 2027

Relevant Coursework: Biomaterials, Biomechanics, Tissue Engineering, Microscopy, Design & Leadership, Organic Chemistry, Transport Phenomena

PROJECTS

Microfluidic Droplet Generator for Artificial Lymph Nodes

Sep. 2026 – Present

BME Design Project, Team Lead

- Leading a 6-person team designing a microfluidic droplet generator to create artificial lymph nodes for studying cancer cell migration, managing the full design-to-fabrication workflow.
- Coordinating SolidWorks device design, analysis of fluorinated oil and collagen flow dynamics, PDMS casting, and soft lithography fabrication.
- Integrating pressure regulation, pump automation, and a cooling system for collagen handling into the device workflow.

TestiCool: Scrotal Cooling Device for Sauna-Associated Fertility Mitigation

Sep. 2025 – Aug. 2026

BME Design Project

- Assumed project leadership in May 2026, reviving a stalled team design effort and driving the project through completion.
- Led design of a sensor-controlled scrotal cooling device targeting heat-induced fertility issues associated with sauna use, independently validating performance at 77°C ambient temperature while maintaining under 15°C surface temperatures.
- Built a Bluetooth mobile app supporting live graphing, three-sensor temperature monitoring, data logging, and historical analysis.

Bone Graft Biomaterials and Bioreactor Design

Jan. 2026 – May 2026

BME Design Project

- Led biomaterial selection for a bone graft system and contributed to bioreactor design and electronics integration.
- Designed a custom sample holder in SolidWorks and manufactured it using manual lathe and mill machining.

EXPERIENCE

Undergraduate Research Assistant

May 2026 – Present

Franck Laboratory, University of Wisconsin-Madison

Madison, WI

- Fabricate PDMS substrates and design experimental fixtures using CAD and 3D printing to support investigation of mechanical strain and substrate stiffness effects on microglial activation, processing approximately 15 samples per week.
- Identified that microglia in monoculture do not appear to respond to high strains and strain rates; directing ongoing investigation toward DAMP exposure, co-culture media from stretched cells, and more physiologically relevant substrate stiffnesses including 1:70 PDMS.
- Execute end-to-end microglial isolations under sterile conditions using primary cell culture, immunocytochemistry, fluorescence microscopy, and ImageJ/Fiji.

Undergraduate Research Assistant

Mar. 2026 – Aug. 2026

Pfleger Laboratory, University of Wisconsin-Madison

Madison, WI

- Co-designed and led fabrication of polymer-based biomimetic root models using ethyl cellulose and PDMS, benchmarked against flat control surfaces to assess bacterial binding capability.
- Quantified bacterial adhesion via CFU assays using an *S. stutzeri* strain and an *E. coli* strain, finding notably greater binding to root mimics than to flat surfaces.
- Performed cloning of a binding gene into plasmid constructs using an *E. coli* strain, executing overnight bacterial culture, PCR, and gel electrophoresis.
- Imaged root mimic substrates using confocal microscopy without staining, resolving surface structures under 10 microns including root hairs and cell wall architecture.

SKILLS

Engineering: SolidWorks, Fusion 360, CAD, 3D Printing, Rapid Prototyping, Microfluidics, Lathe & Mill, Experimental Design

Software/Electronics: MATLAB, COMSOL, Arduino IDE, Bluetooth/BLE, Sensor Integration, PreForm

Lab: Mammalian & Bacterial Cell Culture, PCR, Cloning, Gel Electrophoresis, Immunocytochemistry, Confocal Microscopy, Fluorescence Microscopy, Sterile Technique, Biological Safety Cabinet (BSC), ImageJ/Fiji, PDMS, CFU Assays, Biofilm Assays, Soft Lithography